

SECTION 6

MASS AND BALANCE

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APPENDED TO SECTION 6

Form MBR-1, MASS AND BALANCE RECORD

Form EL-1, EQUIPMENT LIST

SECTION 6

MASS AND BALANCE

6.1 GENERAL

This section provides information required for helicopter loading and computing mass and balance.

It shall be the pilot's responsibility to make certain that:

- the helicopter is properly loaded so that the entire flight is conducted within the Center of Gravity Limits,
- all cargo is stowed and tied down properly so that in-flight shifting is impossible,
- proper tie-down equipment (i.e. ropes, belts, etc.) of sufficient strength has to be used. As many tie-down fittings as possible have to be used per single cargo item in order to reduce the individual load per fitting and to avoid inadvertent in-flight shifting.

6.1.1 Mass Definitions

Basic empty mass consists of the basic helicopter with required standard equipment, optional equipment, unusable fuel, and full operating fluids including transmission, gearbox and engine oils, hydraulic fluid, rotor brake oil, and windshield washer fluid.

Gross mass is the sum of the basic empty mass and the pilot/crew, the passengers, the baggage/cargo, and the fuel. This value will vary with each mission.

6.1.2 Balance Definitions

Reference datum (RD) is an imaginary vertical plane located 3000 mm (118.1 in) forward of the Leveling Point (LP). For balance purposes, all horizontal distances (ARM) are measured from the RD (see Fig. 6-1).

Arm, for balance purposes, is the horizontal distance from the RD to the center of gravity of a given item. For other purposes, Fuselage Stations may be used. Fuselage stations are vertical planes perpendicular to, and measured along, the longitudinal axis of the helicopter. For the BO 105 helicopter arm and fuselage station are the same.

Moment (Massmoment) is the mass of an item multiplied by its arm.

Moment (kgmm) = Mass (kg) x Arm (mm)

Center of Gravity (CG) is the point about which the helicopter would balance if suspended. Distance from the RD is found by dividing the total moment by the gross mass of the helicopter.

$$\text{Arm (mm)} = \frac{\text{Sum of all moments (kgmm)}}{\text{Sum of all masses (kg)}}$$

6.2 BASIC EMPTY MASS CENTER OF GRAVITY

The procedure for establishing mass and moment (relative to the reference datum) of the empty helicopter is described in the BO 105 Maintenance Manual (MM). The MASS AND BALANCE RECORD (Form MBR-1) appended to this Section is used to maintain a continuous history of changes to the basic "as delivered" helicopter mass and balance data.

6.2.1 Mass and Balance Record

The basic empty mass (BEM) and center of gravity (CG) location are determined through actual weighing carried out by the helicopter manufacturer. This data is then entered on the first line of the MASS AND BALANCE RECORD, Form MBR-1 which then becomes a permanent part of the Flight Manual.

The MASS AND BALANCE RECORD must be updated (normally by transcribing the applicable information from the EQUIPMENT LIST) when necessary as follows:

- When additional equipment is installed on the helicopter necessitating a change in the basic empty mass (as per definition), add the new entry or entries to the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).
- Likewise, when equipment is removed from the helicopter, subtract the new entry or entries from the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).

At all times, the last mass and moment entries are considered the current basic empty mass and balance status of the helicopter.

6.2.2 Equipment List

An EQUIPMENT LIST, Form EL-1 is appended to this section and contains optional equipment of the particular helicopter when delivered. Each item on the list is provided with a number and description for identification, together with its mass, arm and moment.

Those items of equipment that were installed when the particular helicopter was initially weighed are so indicated by a check mark (✓) in the "Initial Weighing" column. Therefore the mass, arm and moment of these items are included in the basic empty mass (BEM) data found on the MASS AND BALANCE RECORD, Form MBR-1.

6.3 LOADING DATA (see Fig. 6-1)

The loading data provides location information necessary in calculating the longitudinal CG of the helicopter. Following is a listing of items with their respective arm (fuselage station/location).

6.3.1 Crew and Passenger Stations

	ARM (mm)
Pilot and copilot	1890
Passengers	2690 (2740*)
* Only for Sectioned Bench Seat, FMS 10-21.	

6.3.2 Cargo Tie-down Points

	ARM (mm)
Cabin	2230
	2690
Cargo Compartment	3150
	3380
	3610
	4030

NOTE • Cargo compartment longitudinal center is at ARM 3900mm.

- Ensure that the cargo is properly secured by restraining it from shifting with tie downs.
- Cargo should be centered in the cabin.
- Cargo shall be loaded within the perimeter defined by the tie-down ring centroids.
- The operator shall be responsible for the proper placement and securing of cargo in accordance with standard operating procedures and practices.

6.3.3 Fuel

	ARM (mm)
Supply Tank	2275
Main Tank	3220

6.3.4 Avionics Equipment Bay

ARM = 4250 mm

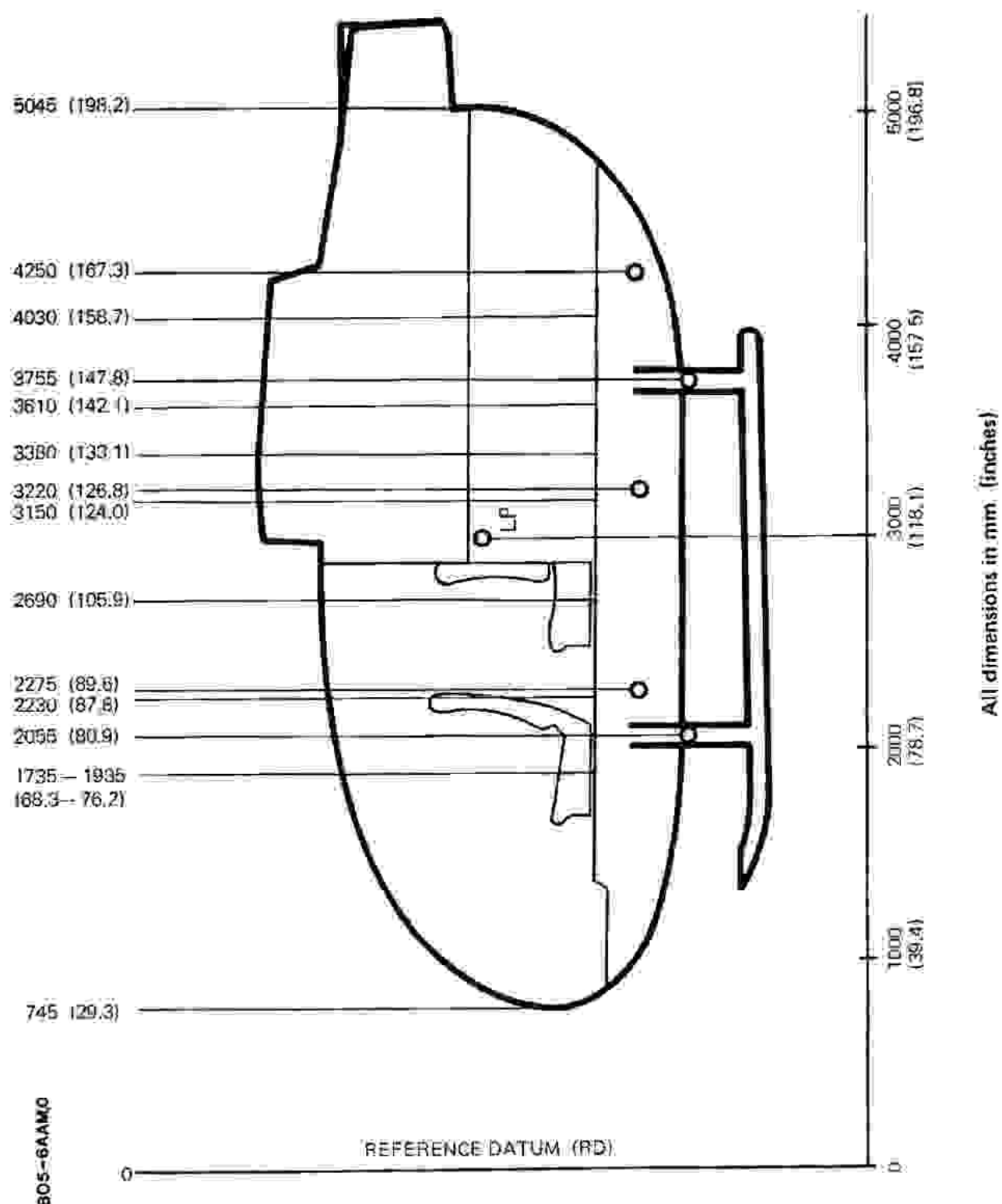


Fig. 6-1 Fuselage Station Diagram

6.4 CALCULATING LONGITUDINAL CG (see Fig. 6-2)

The following examples show a method of calculating the longitudinal center of gravity.

NOTE Basic empty mass data are to be taken from the latest entry on the MASS AND BALANCE RECORD, Form MBR-1 appended to this Section.

LOADING EXAMPLES						
Flight A: 1 pilot (80 kg), bulk cargo load (320 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1270	3360	4267200	1270	3360	4267200
Pilot	80	1890	151200	1350	3273	4418400
Bulk cargo	320	3900	1248000	1670	3393	5666400 ¹⁾
Fuel, supply tank	74	2275	168350	1744	3346	5834750 ²⁾
Fuel, main tank	382	3220	1230040	2126	3323	7064790 ³⁾
Flight B: 2 pilots (2 x 80 kg), 3 passengers (1 x 80 kg, 2 x 90 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1270	3360	4267200	1270	3360	4267200
Pilot and copilot	160	1890	302400	1430	3196	4569600
Passengers (3)	260	2740	712400	1690	3125	5282000 ¹⁾
Fuel, supply tank	74	2275	168350	1764	3090	5450350 ²⁾
Fuel, main tank	382	3220	1230040	2146	3113	6680390 ³⁾
NOTE To make certain that the CG does not "wander" outside of the CG envelope during the flight, the helicopter must be so loaded that the above points ¹⁾ (minimum landing mass), ²⁾ (supply tank full), and ³⁾ (supply and main tanks full) lie within the approved limits.						

NOTE The lateral CG position will always remain within limits regardless of the location of internal loads provided that the gross mass remains within limitations.

6.2 BASIC EMPTY MASS CENTER OF GRAVITY

The procedure for establishing mass and moment (relative to the reference datum) of the empty helicopter is described in the BO 105 Maintenance Manual (MM). The MASS AND BALANCE RECORD (Form MBR-1) appended to this Section is used to maintain a continuous history of changes to the basic "as delivered" helicopter mass and balance data.

6.2.1 Mass and Balance Record

The basic empty mass (BEM) and center of gravity (CG) location are determined through actual weighing carried out by the helicopter manufacturer. This data is then entered on the first line of the MASS AND BALANCE RECORD, Form MBR-1 which then becomes a permanent part of the Flight Manual.

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- When additional equipment is installed on the helicopter necessitating a change in the basic empty mass (as per definition), add the new entry or entries to the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).
- Likewise, when equipment is removed from the helicopter, subtract the new entry or entries from the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).

At all times, the last mass and moment entries are considered the current basic empty mass and balance status of the helicopter.

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An EQUIPMENT LIST, Form EL-1 is appended to this section and contains optional equipment of the particular helicopter when delivered. Each item on the list is provided with a number and description for identification, together with its mass, arm and moment.

Those items of equipment that were installed when the particular helicopter was initially weighed are so indicated by a check mark (✓) in the "Initial Weighing" column. Therefore the mass, arm and moment of these items are included in the basic empty mass (BEM) data found on the MASS AND BALANCE RECORD, Form MBR-1.

6.3 LOADING DATA (see Fig. 6-1)

The loading data provides location information necessary in calculating the longitudinal CG of the helicopter. Following is a listing of items with their respective arm (fuselage station/location).

6.3.1 Crew and Passenger Stations

	ARM (mm)
Pilot and copilot	1650
Passengers	2690 (2740*)
* Only for Sectioned Bench Seat, FMS 10-21.	

6.3.2 Cargo Tie-down Points

	ARM (mm)
Cabin	2230
	2690
Cargo Compartment	3150
	3380
	3610
	4030

NOTE • Cargo compartment longitudinal center is at ARM 3900mm.

- Ensure that the cargo is properly secured by restraining it from shifting with tie downs.
- Cargo should be centered in the cabin.
- Cargo shall be loaded within the perimeter defined by the tie-down ring centroids.
- The operator shall be responsible for the proper placement and securing of cargo in accordance with standard operating procedures and practices.

6.3.3 Fuel

	ARM (mm)
Supply Tank	2275
Main Tank	3220

6.3.4 Avionics Equipment Bay

ARM = 4250 mm

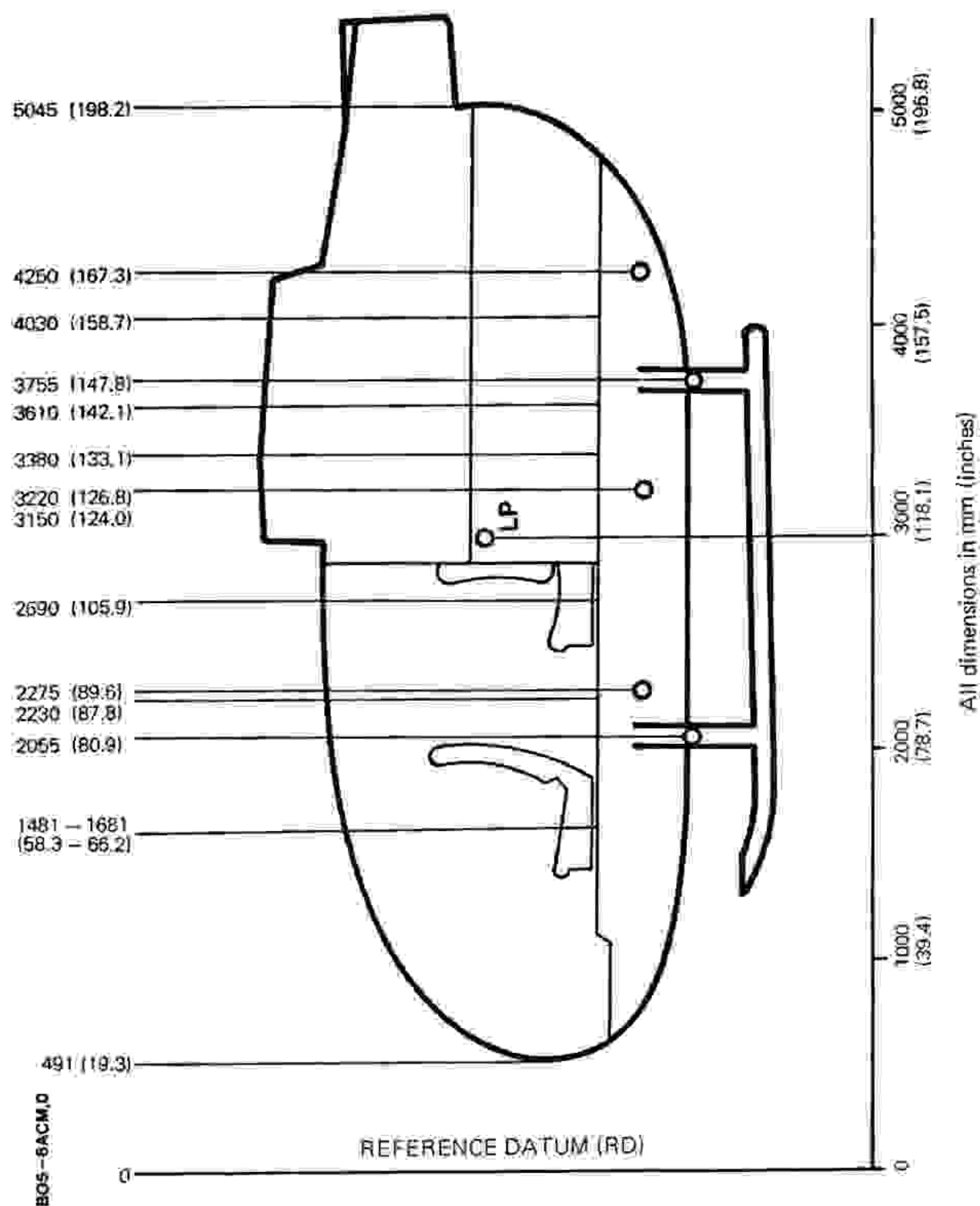


Fig. 6-1 Fuselage Station Diagram

6.4. CALCULATING LONGITUDINAL CG (see Fig. 6-2)

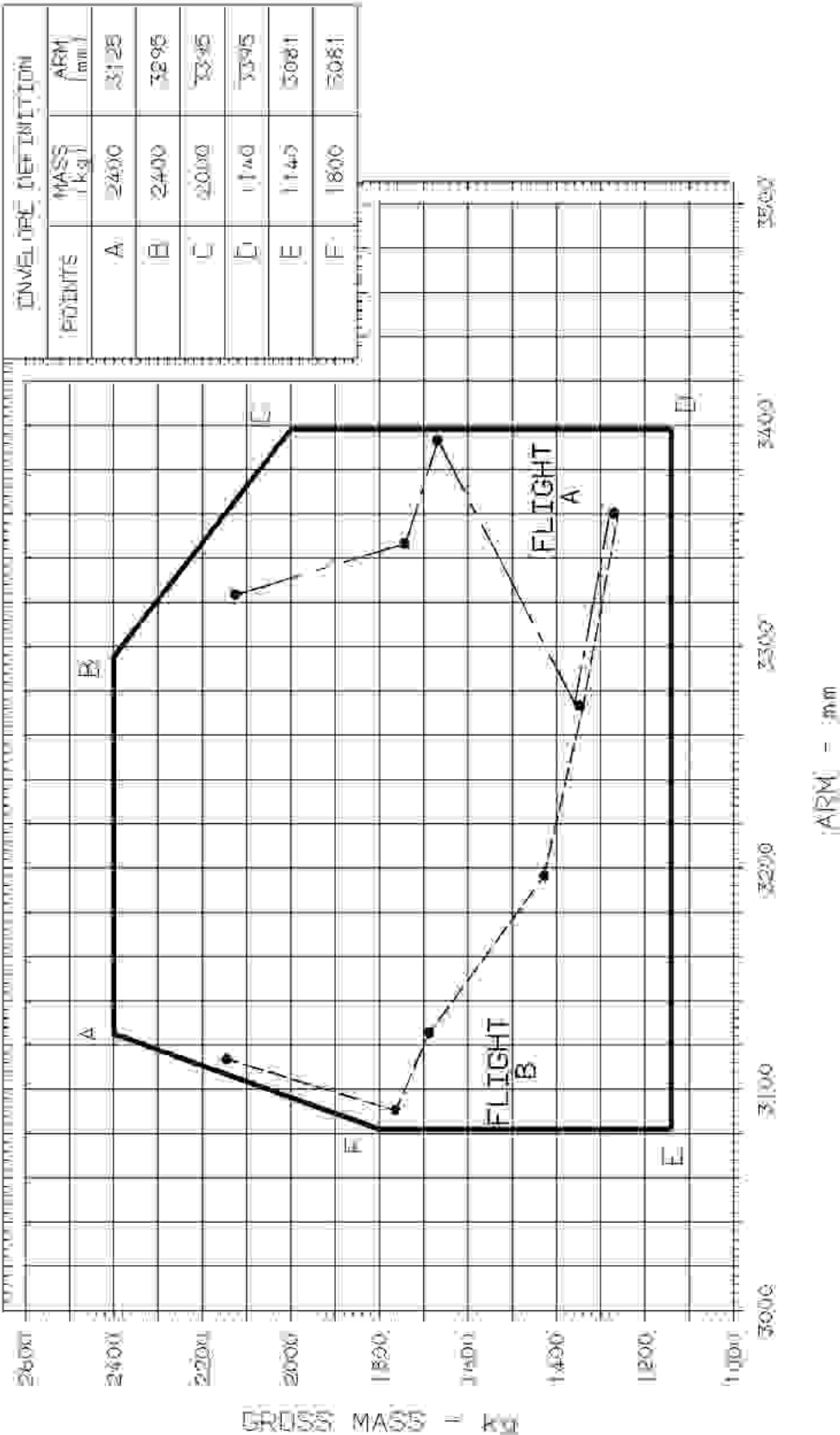
The following examples show a method of calculating the longitudinal center of gravity.

NOTE Basic empty mass data are to be taken from the latest entry on the MASS AND BALANCE RECORD, Form MBR-1 appended to this Section.

LOADING EXAMPLES						
Flight A: 1 pilot (80 kg), bulk cargo load (320 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Resultant					
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1300	3375	4387500	1300	3375	4387500
Pilot	80	1650	132000	1380	3275	4519500
Bulk cargo	320	3900	1248000	1700	3393	5767500 ¹⁾
Fuel, supply tank	74	2275	168350	1744	3346	5935850 ²⁾
Fuel, main tank	382	3220	1230040	2156	3324	7165890 ³⁾
Flight B: 2 pilots (2 x 80 kg), 3 passengers (3 x 80 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Resultant					
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1300	3375	4387500	1300	3375	4387500
Pilot and copilot	160	1650	264000	1460	3186	4651500
Passengers (3)	240	2740	657600	1700	3123	5309100 ¹⁾
Fuel, supply tank	74	2275	168350	1774	3088	5477450 ²⁾
Fuel, main tank	382	3220	1230040	2156	3111	6707490 ³⁾
NOTE To make certain that the CG does not "wander" outside of the CG envelope during the flight, the helicopter must be so loaded that the above points ¹⁾ (minimum landing mass), ²⁾ (supply tank full), and ³⁾ (supply and main tanks full) lie within the approved limits.						

NOTE The lateral CG position will always remain within limits regardless of the location of internal loads provided that the gross mass remains within limitations.

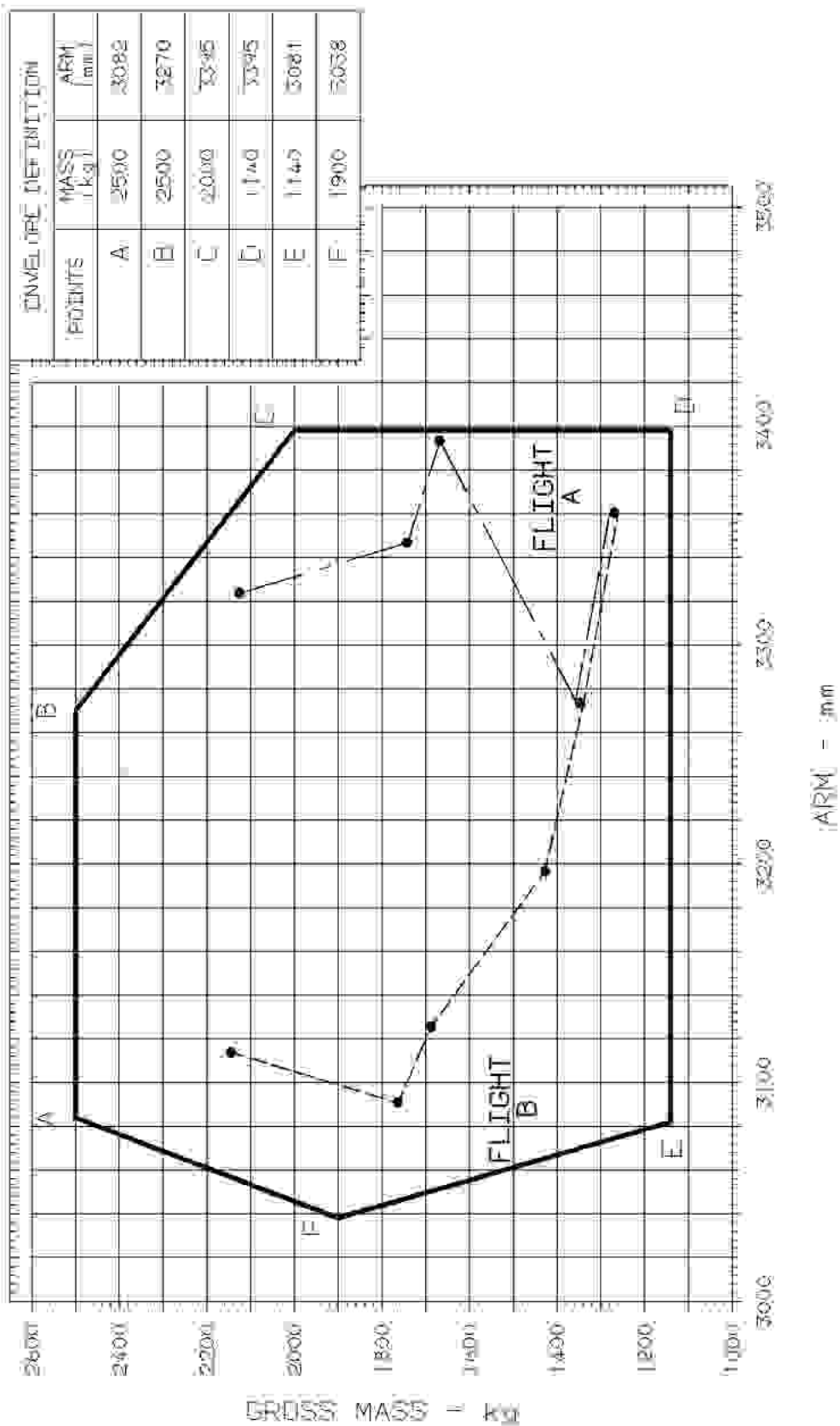
BO5-65EM.1



NOTE REFER TO TEXT FOR LOADING EXAMPLE DATA.

Fig. 6-2 CG Envelope with Loading Examples

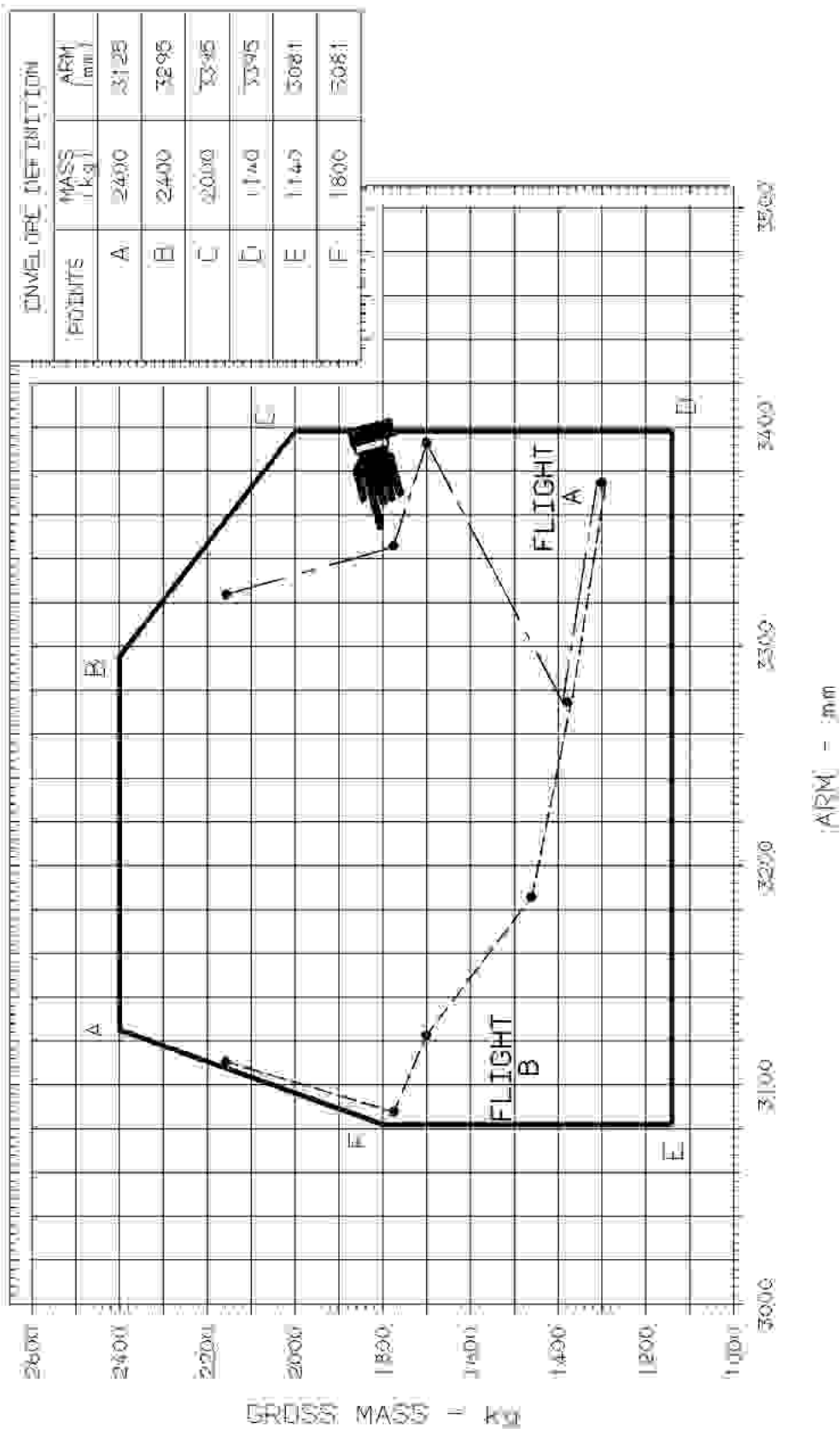
BO5-65GM, 1



NOTE: REFER TO TEXT FOR LOADING EXAMPLE DATA.

Fig. 6-2 CG Envelope with Loading Examples

BO5-65UM-1



NOTE: REFER TO TEXT FOR LOADING EXAMPLE DATA.

Fig. 6-2 CG Envelope with Loading Examples

BO5-651M.1

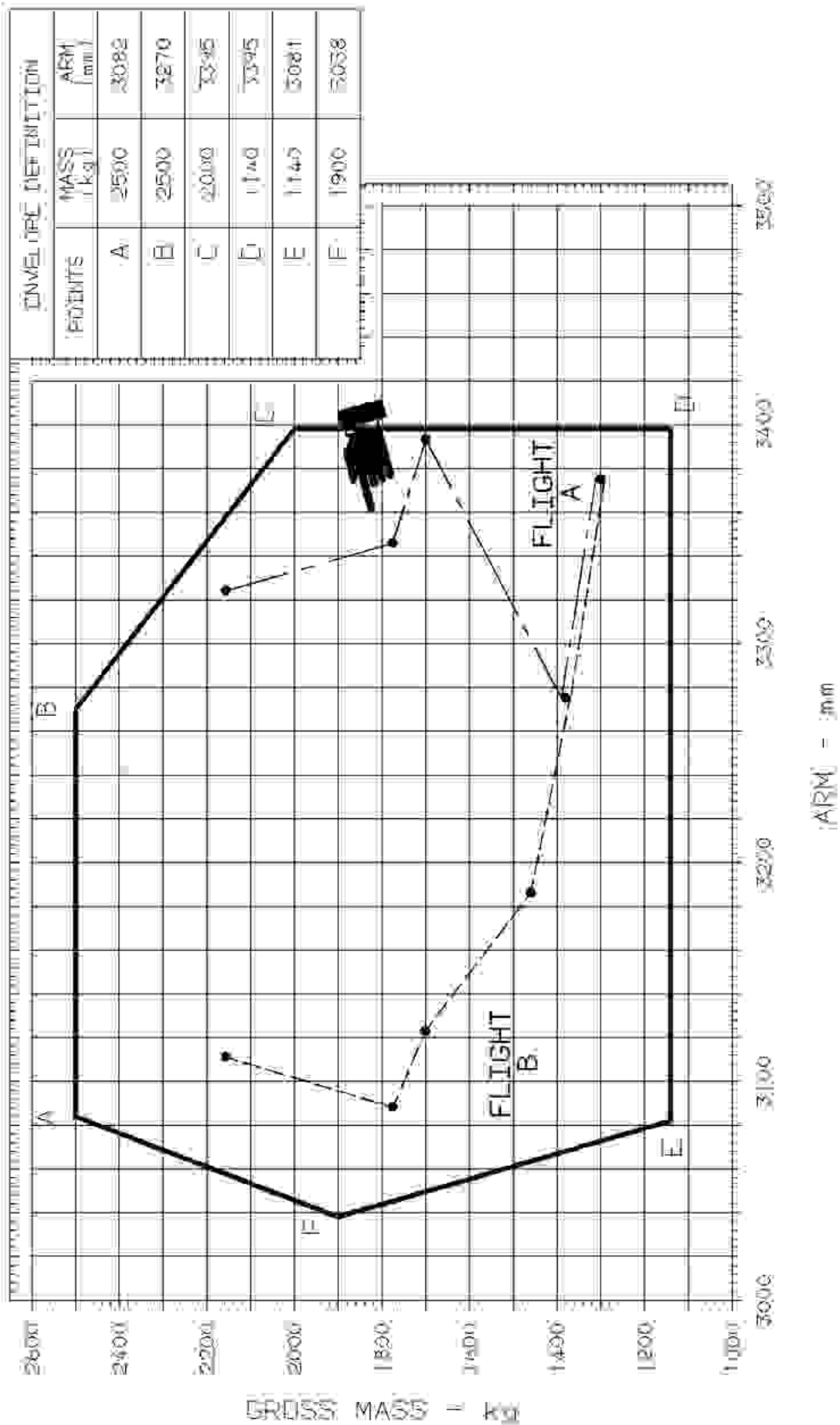


Fig. 6-2 CG Envelope with Loading Examples

BO105 EQUIPMENT LIST (EL)

MODEL/VARIANT:

SERIAL NO.:

Page _____ of _____

[illegible]

1) Required markings: ✓ = installed; 0 = not installed.

